From Data to Solutions · Weekend 7

Counting the FLOPs of a transformer

Where the 6 in 6ND comes from

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Section 1

Where the 6 comes from

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Where the 6 comes from

At 08:00 we used the 6 without deriving it

The one number we did not derive

$$C = \boxed{6}\, N\, D$$

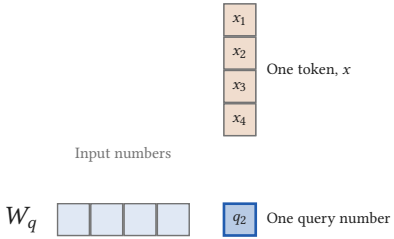
$N = 70 \times 10^9$ parameters

$D = 15 \times 10^{12}$ training tokens

$C = 6.30 \times 10^{24}$ FLOPs

Where the 6 comes from

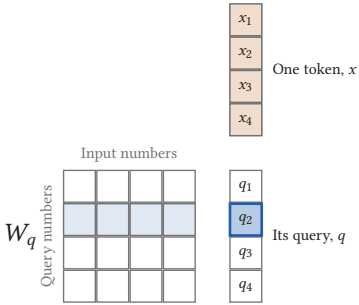
One query number: four multiplies, four adds



- A token is 4 numbers
- One row of W_q is 4 weights, and it makes ONE query number
- Multiply each weight by its number: 4 multiplies
- Add the four products: 4 adds
- 8 FLOP for 4 parameters, so 2 FLOP per parameter

Where the 6 comes from

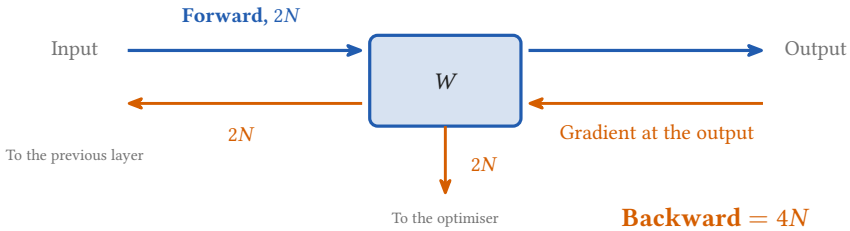
Sixteen weights, sixteen multiply accumulates



- A token is 4 numbers, and its query is 4 numbers
- One query number: 4 multiplies and 4 adds
- All four: 16 multiply accumulates, 32 FLOP
- Two FLOPs per parameter, per token, so the forward pass is $2N$

Where the 6 comes from

The backward pass computes two gradients



- The loss comes back as a gradient at the output
- One matmul sends it on to the previous layer, one turns it into the update for W
- Both are the same shape as the forward, so twice the cost

Where the 6 comes from

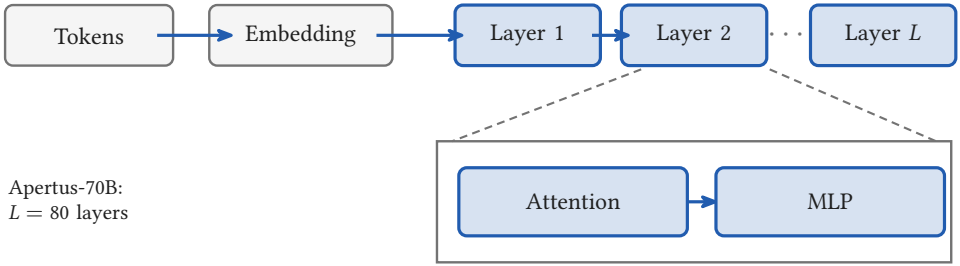
Two plus four is six

$$\text{FLOPs per token} = \underbrace{2N}_{\text{forward}} + \underbrace{4N}_{\text{backward}} = 6N$$

- Over D training tokens, $C = 6ND$: the rule of thumb was a derivation all along

Where the 6 comes from

A transformer is a stack of identical layers



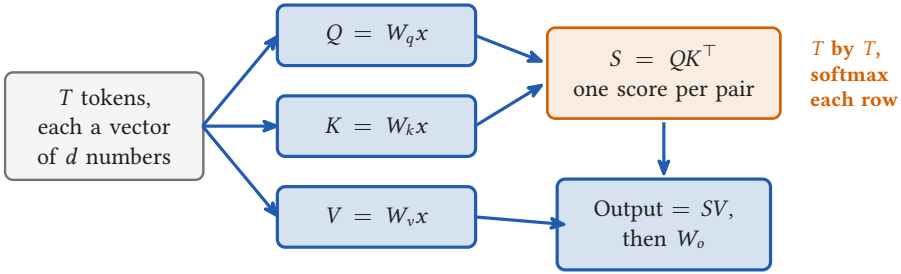
Apertus-70B:
$L = 80$ layers

Every layer, the same two blocks and the same shapes

- Count one layer, multiply by L, and you have the model

Where the 6 comes from

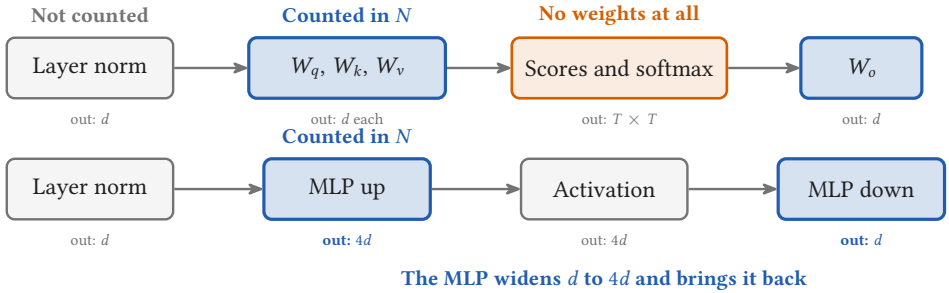
Attention, in one slide



- T is the context length: how many tokens the model reads at once
- Q, K, V and W_o are weight matrices. The scores are not

What the count leaves out

After the embedding, every token is a vector of d numbers. Typically d is a few thousand: Apertus-70B uses $d = 8192$.



Section 2

Counting one layer by hand

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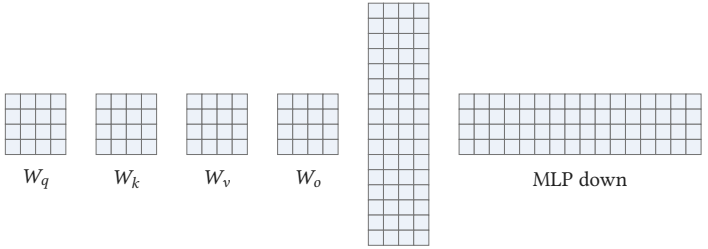
Counting one layer by hand

One layer at $d = 4$: 192 numbers



Attention: $64 = 4d^2$

MLP: $128 = 8d^2$

One layer: $192 = 12d^2$

The vanilla layer: full multi head attention, and a plain MLP that widens d to $4d$ and back. Every cell above is one parameter.

Counting one layer by hand

The same rule on a 2025 model is 6 percent low

- Grouped attention: 8 key and value heads, not 64
- A wider MLP, 5.25 d instead of 4 d, and not gated
- $12d^2 \times 80$ layers gives 6.442×10^{10} against a published 6.845×10^{10}

Counting the real shapes: 70,599,843,840

80 layers, $d = 8192$, 64 heads, 8 key and value heads, head dim 128, hidden 43008, vocab 131072

W_q	8192×8192	67,108,864
W_k	8192×1024	8,388,608
W_v	8192×1024	8,388,608
W_o	8192×8192	67,108,864
Attention $= 2.25\, d^2$		150,994,944
MLP, two of 8192×43008		704,643,072
One layer $= 12.75\, d^2$		**855,638,016**

Times 80 layers	68,451,041,280
Embeddings, untied	2,147,483,648
Norm weights	1,318,912
Total	**70,599,843,840**

Published: 70.60 billion

Counted from `swiss-ai/Apertus-70B-2509/config.json`. Nothing here is asserted.

Section 3

Not all FLOPs are equal

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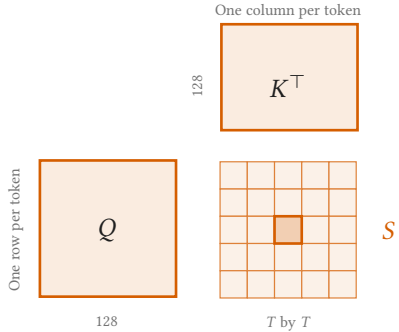
Not all FLOPs are equal

The attention scores have no weights



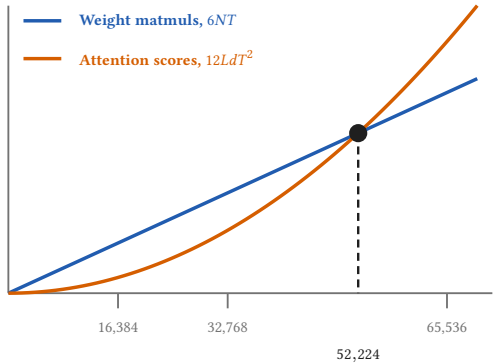
Each cell: 128 multiply accumulates

- One score per pair of tokens, so the grid grows with the context
- No weight matrix anywhere in it, so N never sees it

$$\underbrace{4dT}_{\text{forward}} + \underbrace{8dT}_{\text{backward}} = 12\,d\,T$$

- Per token, per layer, so $12\,L\,d\,T$ for the whole model

Where the two costs cross

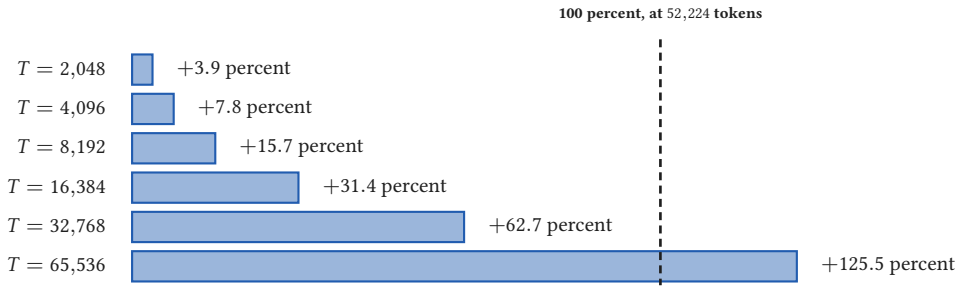


$$T^* = \frac{P_{\text{layer}}}{2d} = 6.375\, d = 52{,}224$$

- Set the two terms equal
- Divide through: what one layer holds is what is left
- That layer holds $12.75\, d^2$, counted a moment ago
- Not a constant of transformers: a vanilla layer crosses at $6d$, a gated one at $8d$

Both attention matmuls counted, no credit taken for causal masking.

What attention adds on top of 6ND



- 4,096 is the sequence length Apertus pretrained at
- 65,536 is the model's own maximum context, where attention more than doubles the bill
- Above the crossover, $6ND$ is the smaller of the two terms